Engineering Project Detailed Research Plan

Please complete the information/questions begun/seen below in red ink. Save your plan to your computer and upload a copy to your application Forms Folder.

Date: **October 3, 2025**
Student Name: **Krishna Bhatt**
Project Details: **Grade 11 Mathematics, Project A99**
Project Title: **THALis: A Topology-Preserving Brain Computer Interface for Dexterous Non-Invasive EEG-Driven Hand Control**

Parts of the generic engineering project are listed below with descriptions to the students in the boxes. Students may provide a detailed research plan by describing their specific project in response to each box below.

Engineering Goal: PROBLEM BEING ADDRESSED: All engineering projects solve a problem or fill a need. This goal should be a simple statement that describes the product being designed, the customer it is for and the problem or need it satisfies. Example" "The goal is to design a solar powered lawn mower for inexpensive automated lawn care for homeowners"

Context: Loss of upper-limb function following amputation poses a profound challenge to independence and quality of life. While prosthetic hands have advanced in mechanical design, current non-invasive neural control systems remain limited: EEG-based interfaces are typically constrained to a few discrete gestures due to low signal-to-noise ratio (SNR), while sEMG-driven systems suffer from noisy surface recordings and limited robustness to electrode shifts. As a result, most users experience only clumsy, coarse manipulation rather than natural, dexterous hand control.

My Project Goal is: The goal of this project is to develop a noninvasive prosthetic hand control algorithm and architecture for fine, dexterous motor control of the hand. This project proposes a new neurocomputational framework for non-invasive prosthetic control that integrates insights from computational neuroscience, nonlinear dynamics, and spiking neural networks (SNNs). The central hypothesis is that the fundamental emergent properties of neural activity—including chaotic dynamics, synchronization, and nonlinear attractor behavior—are partially reflected in scalp EEG, but obscured by low SNR (signal to noise ratio) and conventional linear processing. By developing a novel EEG transformation grounded in chaos and dynamical systems theory, this project aims to amplify these emergent signatures, yielding richer neural features than traditional spectral approaches.

The broader motivation is twofold: (1) to expand the functional capacity of non-invasive prosthetic hands for amputees who cannot access invasive brain–computer interfaces, and (2)

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to demonstrate that computational neuroscience principles—emergent dynamics, SNN plasticity, and multimodal neural fusion—can concretely enhance assistive neurotechnology.

Impact Statistics: This project aims to serve those whose lives have been profoundly affected by hand amputation or loss of motor control of the hand to facilitate reintegration into social and working life.

- Estimates reveal around 3 million people worldwide live with upper limb amputations
- A 2020 study found a 44% rejection rate across all upper-limb amputation levels.
- *Some Activities of Daily Living (ADLs)*: Mobility, eating, dressing, writing, grasping, typing, shoelaces, operating scissors, opening bottles etc.

Design Criteria: Design criteria define the product's required performance . Examples: " It will have a minimum speed of 10 KPH", "The output will be within 15% of the mean of the experimental data". "It must withstand 15 repetitions of a 10N impact" The International System of units (SI) required.

My Project Design Criteria are the following: To pass the threshold of required performance, the control architecture should:

Continuous-Control Quality

1) **Finger individuation index (Degree of independent finger control)**: Fractional dependence of digit movements (with 0 being fully coupled and 1 being fully independent)
    - **Minimum threshold (success)**: ≥ 0.6 across multi-finger tasks, at least 0.2 higher than sEMG-only baseline.

2) **Trajectory RMSE and correlation:** Root-mean-square error (RMSE) between reference (via vision model) trajectories and the prosthetic output; Pearson correlation coefficient
    - **Minimum Thresholds:** Normalized RMSE ≤ 0.15 (i.e., average error ≤15% of full range) and r ≥ 0.85 for primary DOFs (wrist flexion, index finger, etc.) For secondary DOFs, RMSE ≤ 0.25 and r ≥ 0.7.

3) **Latency (end-to-end)**: Time from neural event onset to prosthetic response (ms).
    - **Minimum threshold:** ≤ 200 ms median end-to-end latency.

Information-Theoretic and Signal Metrics

1) **SNR improvement (EEG transform):** Signal-to-noise ratio improvement for control-relevant features after EEG transform (dB).
    - **Minimum Thresholds:** ≥ +6 dB improvement in task-relevant features (equivalent to ~2x amplitude SNR).

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2) **Information Transfer Rate (ITR) or continuous equivalent:** Bits/second representing how much information the system reliably conveys
    - **Minimum Threshold:** ≥ 1.0 bit/s for EEG continuous control in real use.

Additional Metrics: Functional Performance

1) **Task success rate (gross and fine manipulation)**: Percent of tasks completed successfully (e.g., Box-and-blocks, tying shoelaces, writing, clothespin relocation, etc.)
    - **Minimum threshold (success):** ≥ 80% success on standard gross tasks and ≥ 65% success on fine manipulation tasks; must exceed sEMG-only baseline by ≥ 20 percentage points

2) **Dexterity / kinematics (Path smoothness & jerk)**: Path length and normalized integrated jerk of the fingers during tasks (lower jerk means smoother movement)
    - **Minimum threshold (success)**: Normalized jerk reduction ≥ 25% relative to sEMG-only controller; path length within ±20% of reference smooth trajectory

NOTE: The functional testing of the algorithm will be conducted with an open-source 3D printed robot arm (files provided here: https://www.dexhand.org/) and OpenBCI EEG headcap (https://docs.openbci.com/AddOns/Headwear/MarkIV/), if permitted by time constraints prior to the fair. The mechanical designs are not a core component of my project. Images from the respective sites are provided below.

Diagnostic/Internal Measurements

1) **EEG Channel Diagnostics:** Channel impedance, RMS baseline noise, percentage of rejected channels per session.
2) **Transform separability:** Class separability for discrete actions (Fisher score, Cohen's, or between-class variance ratio on transformed features)
3) **Nonlinear dynamics diagnostics:** Computed dynamical features (Permutation entropy, sample entropy, recurrence determinism, estimated largest Lyapunov exponent during task vs. rest)
4) **SNN Internal Metrics:** Mean firing rates, firing-rate variance, synaptic weight distributions, convergence indicators, etc.
5) **Classifier/decoder performance diagnostics:** Confusion matrix, regression residuals and Bland-Altman plots for continuous decoders.

Constraints: Constraints are factors that limit the engineer's flexibility such as size, cost, and time limitations. Examples: "It must fit in a box no larger than 10x20x50 cm" "The maximum cost is $50" "The software must run in real time on a Raspberry Pi"

My Project Constraints are the following:

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- **Hardware–** Software development, compilation, and testing must be completed on my home personal machine (GTX 1080 Ti, 16GB RAM, Ryzen 2700x) and Arduino UNO breakout board
- **Budget–** The maximum budget of the project on materials is $300 to be allocated to 3D Printing materials for hardware prototype/demonstration. Software must be compiled, tested, and run on home materials.
- **Tools–** All software and development tools must be open-source and free to use or under education plan.
- **Time**– I have limited time for development each week (around ~35-40 hours) until the fair in March.

Provide your chosen design. For hardware, provide a sketch. For software, provide a flowchart. Indicate the components you will develop, and the libraries you are using.

My Project Design is shown below: insert photos, diagrams, or illustrations below.

My project will be built to meet four key objectives:

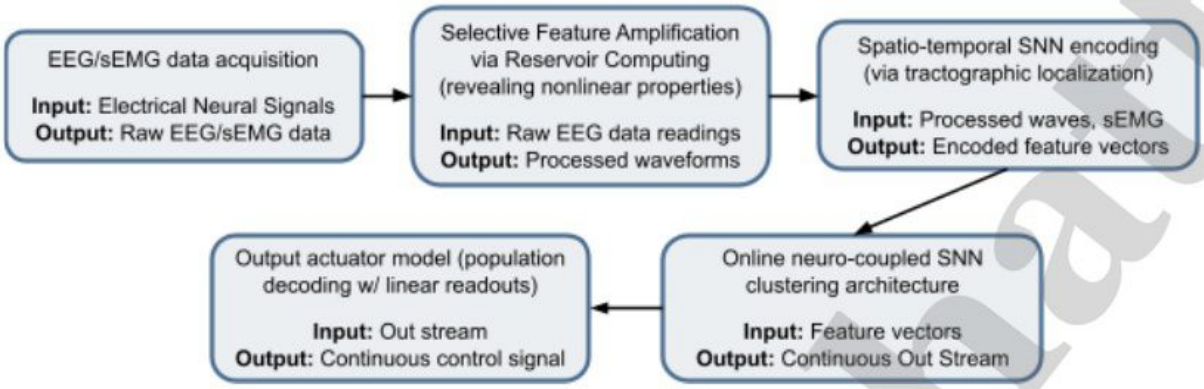
(1) Decomposition and elucidation of nonlinear neural dynamics within the context of electrical signal generation and competing white/grey matter regions.
(2) Develop a high-dimensional EEG signal representation using tractography-weighted reservoir computing, amplifying nonlinear emergent neural dynamics (oscillatory modes, synchrony, chaos) and improving SNR.
(3) Design and implement a spiking neural network (SNN) for online prosthetic control, integrating EEG-transformed signals with sEMG features to form a probabilistic and continuous model of user intent.
(4) Rapid prototyping of EEG / sEMG with computer simulation, 3D printing technology?

OVERVIEW AND TIMELINE

An iterative approach: I plan to go about this project by building one abstraction at a time, focusing on various of the four major components. Each part will be developed modularly and specifically with the intention of massively augmenting prosthetic dexterity through only commonly sourced EEG tools. I have currently researched quite a bit into existing solutions and other products (cited below) and these are my initial ideas about how each component would likely function; however, they are likely to change.

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Flowchart of Control Algorithm



The guiding principle behind this design is the motivation to make the process of utilizing a prosthetic similar to utilizing a previous hand. Thus, this architecture exploits operating neural dynamics and emergent properties from constrained brain regions to amplify the signal-to-noise ratio and guide SNN behavior which decodes activity.

PROBLEM FRAMING

To tackle the issue of inferring true signal from noise in EEG data readouts, I plan to frame the production of such electrical signals as a linear output from a dynamical/bifurcational system of neural connections. Here, I would build upon current findings from theoretical neuroscience (thresholding, bursting, refractoriness, synchronization etc.) to elucidate the specific emergent behaviors (saddles, attractors, oscillations) within key regions of the brain. Here are a couple of concepts that I would be using.

Neural Dynamics: To model neural activity, one can span a spectrum from detailed biophysical models to simplified phenomenological models. Two common models that I would likely incorporate to describe neuronal behavior would be the Hodgkin-Huxley Model or the Izhikevich model. The Hodgkin-Huxley model is the gold standard for biophysical models and describes action potentials arising from ionic currents.

$$C_m \frac{dV}{dt} = -g_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - g_{\mathrm{K}} n^4 (V - E_{\mathrm{K}}) - g_L (V - E_L) + I_{\mathrm{ext}}$$

Each gating variable (*m, h, n*) follows the first-order kinetics:

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x, \quad x \in \{m, h, n\}$$

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This provides a biophysically faithful description of neurons, however incurs a significant computational cost. To capture diverse firing behaviors with minimal overhead, the Izhikevich model is commonly used by reducing the system to two coupled equations.

$$\begin{aligned} \frac{dv}{dt} &= 0.04v^2 + 5v + 140 - u + I, \\ \frac{du}{dt} &= a(bv - u), \\ \text{if } v \geq 30\ \text{mV}, &\quad v \leftarrow c, \quad u \leftarrow u + d \end{aligned}$$

This simpler form can mimic tonic spiking, bursting, and fast spiking, which could potentially make it a valuable model for network-scale simulations. Both HH and Izhikevich models can be analyzed in terms of phase portraits of system trajectories in state space. Here, I could observe certain patterns such as fixed points, limit cycles, and chaotic attractors. These features correspond directly to neural states utilized to derive patterns of observed electrical signals. Additionally, empirical studies show that white matter constrains functional activity most present in grey matter and thus can be used to weight reservoir connectivity, embedding anatomical priors.

Reservoir Computing: While traditional artificial neural networks require extensive training, reservoir computing uses the intrinsic dynamics of a recurrent network to project input into a high dimensional space, making it optimal for a selective feature transform of the raw EEG data. Only the output layer is trained, making it sensitive to internal neural dynamics with powerful temporal processing and computationally efficient. The structure mirrors the brain's cortical microcircuits, including recurrent networks of excitatory and inhibitory neurons with rich dynamics, where downstream regions extract functional outputs. A reservoir computer updates its internal state *x(t)* based on incoming signals *u(t)*, providing an output *y(t)*:

$$\mathbf{x}(t+1) = f(W_{in}\mathbf{u}(t) + W\mathbf{x}(t))$$

$$\mathbf{y}(t) = W_{out}\mathbf{x}(t)$$

Key components:

- W_{in}: input-to reservoir weights (mapping EEG signals into network)
- W: recurrent weight matrix (structured using white-matter tractography)
- W_{out}: trainable output weights (mapping to prosthetic commands)

The echo state property ensures that reservoir states asymptotically depend on the input history rather than arbitrary initial conditions; a common sufficient condition would be $\varrho(W)$ <

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1, where $\varrho(W)$ is the spectral radius of the reservoir's recurrent weight matrix. This stabilizes dynamics while preserving nonlinear richness.

Dynamics and Bifurcation Theory: Formally, a nonlinear dynamical system would be defined as $\dot{x} = f(x, p)$ where the trajectory $x(t)$ depends on both the current state and a set of tunable parameters. For neural systems, x would represent voltages/channel activations, while p corresponds to conductances/synaptic strengths. When parameters shift gradually in this system, the neural system would undergo bifurcations, sudden qualitative changes in activity.

- *Hopf Bifurcation:* produces rhythmic oscillations (seen in cortical oscillations & motor rhythms)
- *Saddle-Node Bifurcation:* underlies the "all-or-none" property of action potentials
- *Period-doubling cascades*: can drive neurons into chaotic regimes

These bifurcations allow a single neural system to switch between functional regimes (resting, spiking, bursting) based on context, mirroring how cognition adapts to task demands. Mathematically, bifurcations occur when the eigenvalues of the Jacobian matrix cross critical boundaries:

$$\mathbf{J} = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\mathbf{x}^*}, \quad \lambda(\mathbf{J}) \in \mathbb{C}$$

For a Hopf Bifurcation:

$$\mathrm{Re}(\lambda) = 0, \quad \mathrm{Im}(\lambda) \neq 0$$

This connects local stability analysis directly to emergent global phenomena. At certain operating points, nonlinear systems enter chaotic regimes, where small perturbations in inputs produce diverging trajectories. The degree of this chaos can be quantified by the Lyapunov exponent, defined as

$$\lambda_{\max} = \lim_{t \to \infty} \frac{1}{t} \ln \frac{||\delta \mathbf{x}(t)||}{||\delta \mathbf{x}(0)||}$$

if $\lambda_{max} > 0$, trajectories would diverge exponentially, signalling chaos. In my instance of reservoir computing chaos can be exploited to provide the system with high-dimensional, flexible activity patterns which can be harnessed for computation.

MODEL DEVELOPMENT AND HARDWARE

The control architecture will be developed using Python in conjunction with the Brian2 Neuron simulator package and the Brian2CUDA package for GPU parallelization. All model training and

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testing will be conducted using EEG/sEMG data collected with the g.tec Unicorn Hybrid EEG headset operating on myself (Krishna Bhatt).

- *Machine specifications:* Ubuntu Jammy Jellyfish, GTX 1080 Ti, 16GB RAM, Ryzen 2700x
- *EEG specifications:* g.tec Unicorn Hybrid EEG Headset, 8 channel
- *Simulation*: The prosthetic arm would be simulated/visualized with PyBullet Physical Simulator
- *Robotic Arm*: 3D printed Dexhand V1 (construction if time permits, accessory to project)

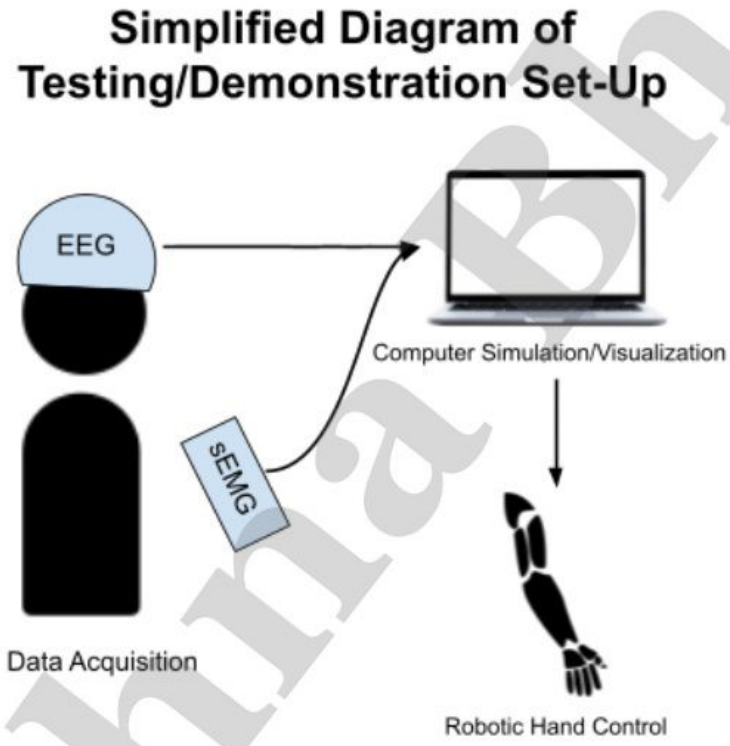


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Test and evaluate your prototypes against the design criteria listed above to show how well the product meets the need/goal. Provide a test plan describing how you will test the design criteria and constraints you listed above., How will you analyze the data? If the product requires human testing please fill out and append https://science-fair.org/wp/wp-content/uploads/2015/10/Research-Plan-Human-Participants.docx

I test and analyze my prototypes using the following methods: In order to test the designed control architecture and to evaluate it against the current standard methods (simple myoelectric) and user intent, there are two control groups: (1) a purely sEMG based control

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algorithm (https://doi.org/10.3389/fnins.2021.621885); (2) a computer vision model tracking user intent. These two controls allow me to compare the effectiveness of the EEG data integration methodology in addition to testing accuracy of determined user intent in a simulation of the arm. Functional characteristics would be tested with a hardware prototype (again, time dependent). I have listed below how I will evaluate each of the design criteria.

Finger individuation index	Quantifies the degree of independence in controlling each finger. Calculated from correlations between finger movement trajectories — higher values mean more independent, dexterous finger control. $$\text{Individuation Index} = 1 - \frac{\text{Var}(F_{\text{ens}})}{\sum_{i=1}^{n} \text{Var}(F_i)}$$ where F_{ens} = ensemble finger motion, F_i = individual finger trajectories, and n = number of fingers. Values range 0 (fully coupled) to 1 (fully independent).
Trajectory RMSE and correlation	Measures how closely the prosthetic's trajectory follows a target trajectory (RMSE) and the similarity of temporal patterns (correlation). $$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(x_{\text{pred}}(t) - x_{\text{true}}(t)\right)^2}$$ $$r = \frac{\sum_t (x_{\text{pred}}(t) - \bar{x}_{\text{pred}})(x_{\text{true}}(t) - \bar{x}_{\text{true}})}{\sqrt{\sum_t (x_{\text{pred}}(t) - \bar{x}_{\text{pred}})^2 \sum_t (x_{\text{true}}(t) - \bar{x}_{\text{true}})^2}}$$ where $x_{pred}(t)$ = prosthetic output, $x_{true}(t)$ = target trajectory.
Latency (end-to-end)	Time delay between onset of neural/muscle activity and the corresponding prosthetic motion. Critical for closed-loop usability. $$\text{Latency} = t_{\text{response}} - t_{\text{onset}}$$ where t_{onset} is the neural/sEMG event start and $t_{response}$ is actuator initiation time.
SNR improvement (EEG transform)	Improvement in signal-to-noise ratio (SNR) of task-relevant EEG features after applying the nonlinear transformation compared to raw EEG.

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	$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right)$$ $$\Delta\mathrm{SNR} = \mathrm{SNR}_{\mathrm{after}} - \mathrm{SNR}_{\mathrm{before}}$$ where P_{signal} = variance of power in task-related bands/features, P_{noise} = variance in unrelated frequencies or baseline.
Information Transfer Rate (ITR)	Measures efficiency of the system in transmitting user intent, in bits per minute. Higher ITR means more information conveyed from neural activity to control. $$\mathrm{ITR} = \left[\log_2 M + P\log_2 P + (1-P)\log_2\left(\frac{1-P}{M-1}\right)\right] \times \frac{60}{T}$$ where M = number of possible targets, P = classification accuracy, T = average trial time (s).
Task success rate	Percent of task attempts successfully completed (e.g., object transferred without drop). Reflects functional usability.
Dexterity	Smoothness and efficiency of movements; quantified via jerk (time derivative of acceleration) or path efficiency. Lower jerk and shorter path length relative to optimal indicate higher dexterity. $$\mathrm{NJ} = \frac{1}{T^2 A^2}\int_0^T \left(\frac{d^3x(t)}{dt^3}\right)^2 dt$$ where T = movement time, A = amplitude of movement. Lower normalized jerk means smoother, more dexterous control.

Human Participants Research:

(1) *Participants:* Describe age range, gender, racial/ethnic composition of participants. Identify vulnerable populations (minors, pregnant women, prisoners, mentally disabled or economically disadvantaged

(2) *Recruitment:* Where will you find your participants? How will they be invited to participate?

(3) *Methods:* What will participants be asked to do? Will you use any surveys, questionnaires or tests? If yes and not your own, how did you obtain? Did it require permissions? If so, explain. What is the frequency and length of time involved for

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each subject?

(4) *Risk Assessment:* What are the risks or potential discomforts (physical, psychological, time involved, social, legal, etc.) to participants? How will you minimize risks? List any benefits to society or participants.

(5) *Protection of Privacy*: Will identifiable information (e.g., names, telephone numbers, birth dates, email addresses) be collected? Will data be confidential/anonymous? If anonymous, describe how the data will be collected. If not anonymous, what procedures are in place for safeguarding confidentiality? Where will data be stored? Who will have access to the data? What will you do with the data after the study?

(6) *Informed Consent Process*: Describe how you will inform participants about the purpose of the study, what they will be asked to do, that their participation is voluntary and they have the right to stop at any time.

(1) Participants: This project (including model training, testing, and deployment) will be conducted personally on the researcher (Krishna Bhatt).

- *Description:* Age 16, Male, Indian-American

(2) Recruitment: I will be the only one participating in the experiment

(3) Methods: I will have to utilize the EEG and sEMG devices in performing various predefined muscle movements with my hand, recording all of the output data. These tasks can range from individual finger movement to grasping or other gestures. Usage time may take from 3-8 hours through trial and error and testing.

(4) Risk Assessment: This project employs non-invasive dry electroencephalography (EEG) and surface electromyography (sEMG), both of which are widely used in neuroscience and clinical research with well-established safety records. This research would ultimately help develop more dexterous non-invasive prosthetics which would have a great social impact to amputees.

- *EEG Safety*: EEG uses passive scalp electrodes to measure naturally occurring brain signals. No electrical current is introduced into the body; only microvolt-level potentials are recorded. Risks are negligible and limited to potential minor skin irritation at electrode contact points.
  - GTEC Unicorn Hybrid EEG Safety Notice & Docs: https://github.com/unicorn-bi/Unicorn-Suite-Hybrid-Black-User-Manual/blob/main/README.md#safety-notice
- *sEMG Safety:* sEMG sensors record muscle activity from the surface of the skin using adhesive or dry electrodes. Like EEG, no current is applied, and risks are negligible, aside from rare mild skin irritation.
  - uMyo sEMG Docs: https://make.udevices.io/guides/umyo-quick-start/

(5) Protection of Privacy: I will be testing on myself, but no data will be published (all EEG and EMG data will remain anonymous). The data would be collected through long running sessions using the two devices for various movements and will be stored on my

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personal machine, where only I will have access to my own data. This data will later be used for training/testing my project.

(6) Informed Consent Process: I will be managing the use of the EEG/sEMG devices on myself and would have the ability to stop at any time.

Bibliography: List at least five (5) major references (e.g. science journal articles, books, internet sites & dates of review) from your literature review. If you plan to use vertebrate animals, one of these references must be an animal care reference.
Example: Author 's Name, Year of publication, "Quoted Title of Magazine Article (magazines only)"; Underlined *Title of Book or Magazine, date, volume, and number of magazine issue. Page numbers read.*
If you use a web site: www.urlname.ext, name of topic from the home page, author, and date read.

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